

Section A: Pure Mathematics [40 marks]

- 1** A curve is given by the equation $y = e^{x^2+1} - 2x - 2$. Find
- (i) the derivative of y , [1]
 - (ii) the coordinates of the point of intersection of the graphs of y and its derivative, giving your answer to 3 decimal places, [2]
 - (iii) the equation of the normal to the curve $y = e^{x^2+1} - 2x - 2$ at the point of intersection found in part (ii). [2]

- 2** The functions g and hg are defined by

$$g : x \rightarrow \ln x, \quad x \in \mathfrak{R}, x > 0,$$

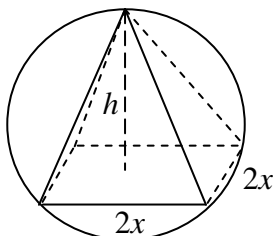
$$hg : x \rightarrow (\ln x)^2 + \ln \frac{e^2}{x^2}, \quad x \in \mathfrak{R}, x > 0.$$

- (i) Find the function h , stating its domain and range. [4]
 - (ii) Explain briefly why the inverse function h^{-1} does not exist. [1]
 - (iii) The function h has an inverse if its domain is restricted to $x \leq k$. State the largest value of k . Hence, find the inverse function h^{-1} . [4]
- 3** (a) The region R is bounded by the curve $y = x^3 + 3x^2 - 2x + 3$ and the line $y = 4x + 11$ for $-2 \leq x \leq 1$. Show this region R clearly on a sketch. Hence, find the area of this region R . [6]
- (b) Find the following integrals:
- (i) $\int \left(\frac{3}{(2x+1)^2} - \frac{1}{\sqrt{e^{x+3}}} \right) dx,$
 - (ii) $\int \frac{3x^2 - x + 4}{3x - 1} dx.$ [6]

- 4** Without using a calculator, solve the simultaneous equations

$$y = 2^{3-2x} + 3 \text{ and } y = \frac{17}{2^x} + 2^{4-x} - 1. \quad [4]$$

- 5 (a) The hull of a submarine is built to withstand extreme pressure, P units, with increasing submerged depth D km. Given that $P = \frac{3}{D^3} + \ln(2D+1)^2$ and the rate of increase of D is 4 km/h, find the rate of increase of P when D is 2 km. [3]
- (b) A square base pyramid of height h cm and length $2x$ cm is inscribed in a sphere of radius 6 cm as shown below. The volume of the pyramid is denoted by $V \text{ cm}^3$.



- (i) Show that $V = 8x^2 + \sqrt{64x^4 - \frac{32}{9}x^6}$. [3]
- (ii) Use differentiation to find the value of x for which V has a maximum value. Hence, find the corresponding maximum value of V . [4]
- [Volume of pyramid = $\frac{1}{3} \times \text{base area} \times \text{height}$]

Section B: Statistics [60 marks]

- 6 In the field of stem cell research, it is known that for every 10 experiments performed by a scientist, only 1 of them is successful.
- (i) Find the probability that of 15 experiments performed, at least 11 of them are not successful. [2]
- (ii) Using a suitable approximation, find the probability that out of 1500 experiments, less than 125 of them are successful. [3]
- (iii) A scientist conducts 25 experiments daily over a large number of n randomly chosen days. Let $\bar{Y}$ be the mean number of successful experiments performed. Find the least value of n for $P(\bar{Y} < 2) \leq 0.01$. [4]
- 7 A company is commissioned by National Environment Agency to find out how the recent “Save Energy” campaign has influenced the lifestyle of Singaporeans. One of the surveyors wants to contact 100 households using a list of numbers in the residential phone directory. He randomly selects one from the first 10 numbers to call and then selects every 10th number after the first selected number.
- (i) What type of sampling method is being described above? [1]
- (ii) State one advantage and one disadvantage of the sampling method used in this context. [2]
- (iii) If the ratio of 3-room, 4-room and 5-room flats in a neighbourhood is 3: 5: 2 respectively, how should the surveyor select a stratified sample of 200 households to interview? [2]

- 8** The lifespan, in years, of a randomly chosen incandescent light bulb is normally distributed with mean 1.5 and standard deviation 0.5. On the other hand, the lifespan, in years, of a randomly chosen fluorescent tube is normally distributed with mean 2.6 and standard deviation 0.2.
- Find the probability that two randomly chosen fluorescent tubes have a total lifespan between 18 to 57.6 months. [2]
 - Two fluorescent tubes are randomly chosen. Find the probability that one has a lifespan longer than 2.6 years and the other has a lifespan less than 3 years. [3]
 - Find the probability that a box of 20 randomly chosen incandescent light bulbs contains at most 14 light bulbs which have lifespan each less than 2 years. [3]
 - Find the probability that the lifespan of a randomly chosen fluorescent tube is more than twice the lifespan of a randomly chosen incandescent light bulb. [3]
- 9**
- In a particular year, it was found that, on average, 3 red abalone in 100, harvested by a diver, had a shell length of less than 55 mm. Assuming that the shell length of a red abalone has a normal distribution with mean 80 mm and standard deviation σ mm, find the value of σ , correct to 2 decimal places. [3]
 - Based on previous records, the shell length of a blacklip abalone harvested in Tasmania is normally distributed with mean 83 mm and standard deviation 16 mm. Before processing, each blacklip abalone at a factory is classified as one of the three grades *A*, *B* and *C* based on its shell length, with grades *A* and *C* having the longest and shortest shell length respectively. It is given that an abalone is classified as grade *C* if it has a shell length of at most 79 mm. Find the minimum shell length of a grade *A* abalone if the proportion of grade *A* abalone is half that of the grade *B* abalone. [3]
- In a particular batch delivered to the factory, 250 blacklip abalone were classified as grade *C*. Estimate the number of blacklip abalone in the batch. [1]
- 10** For ten consecutive months, a factory manager recorded the number of items (in thousands) produced by the factory, y and the total number of workers, x . The manager's data is summarized in the following table:

Number of workers, x	19	25	30	39	45	59	63	72	87	95
Number of items (in thousands), y	29	29	45	90	68	114	134	152	178	168

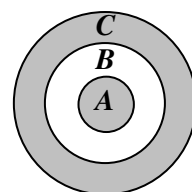
- Give a sketch of the scatter diagram for the data. [2]
- Calculate the product moment correlation coefficient. [1]
- Find the equation of the regression line of y on x as well as the equation of the regression line of x on y . [2]
- Use the appropriate regression line to estimate the number of items (in thousands) produced by 120 workers. Comment on the reliability of this estimate. [3]

- 11** The aviation authorities investigated the average transit time of 250 randomly chosen passengers at the airport. The transit time, x hours, of each passenger is summarized as follows:

$$\sum (x - 4) = -106 \text{ and } \sum (x - 4)^2 = 1104$$

- (i) Test, at the 2% significance level, whether the mean transit time of a passenger is longer than 3.4 hours. [6]
- (ii) The same sample is used to test the null hypothesis $\mu = \mu_0$ against the alternative hypothesis $\mu \neq \mu_0$ at 1% significance level, where μ denotes the population mean transit time of a passenger. Find the set of possible values of μ_0 if the null hypothesis is rejected. [3]
- 12** (a) “It is given that $P(A) \neq 0$ and $P(B) \neq 0$. If events A and B are independent, then A and B are mutually exclusive.”
State with a reason whether the above statement is true or false. [1]
- (b) In a game at a carnival, Roger throws two darts at a dartboard as shown below. The following table indicates the respective area and points awarded for the 3 regions A , B and C . No point will be awarded for not hitting the dartboard. It is assumed that the dart will not hit exactly on the boundary of the two circles.

Region	Area of region (units ²)	Points awarded
A	1	7
B	4	5
C	9	2



It is given that both throws are independent and the darts will hit the dartboard.

- (i) Find the probability that he scores a total of 14 points. [1]
- (ii) Roger wins a prize if he scores at least 12 points. Given that the probability of winning at least 1 prize in n games is greater than 0.6, find the least possible value of n . [4]

Jimmy also throws two darts at the same dartboard with his first dart hitting the dartboard. For his second throw, the probability of hitting the dartboard is

- 0.6 if he obtains more than 2 points for the first throw,
 - 0.3 if he obtains exactly 2 points for the first throw.
- (iii) Construct a tree diagram showing all possible outcomes for the 2 throws. [2]
- (iv) If Jimmy scores a total of 7 points, find the exact probability that he scores 5 points in his first throw. [3]

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